CHEMICAL SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture

DBA-CHEM: Chemical Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting chemical-sector critical infrastructure across production, transport, distribution, and enterprise-control functions.

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Chemical sector. It is the parent instrument for the DBA-CHEM series and governs the facility-class instruments of every type developed beneath it. Like every framework in the Design Basis for AI Governance in Critical Infrastructure, it applies the DB → DBA → DBA-MA three-layer analytical progression: a baseline Design Basis (DB) establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a Military-Action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events.

The distinguishing architectural choice of this framework is that it organizes the chemical sector by function rather than by substance. The rationale and the resulting taxonomy are established in Section 1.3 and developed in Section 3.

1.2 What This Document Does Not Do

This document does not supersede or modify any existing chemical-sector regulation or standard. It does not restate the governing definitions of the Bright Line or the Command Broker; those are inherited by reference from FD-BL and its determination annexes, and this framework develops only the sector-specific placement and Command-Broker application for chemical-sector facilities. It does not, at this version, develop the per-class analytical content: the facility-class instruments of every type named in the register (Section 8) carry that work.

1.3 Function-Primary Organizing Principle

The chemical sector is unusual among NIPP sectors in that it is defined by a material rather than by a service. The electric, water, and natural-gas sectors are each defined by a function they perform — deliver power, deliver potable water, deliver gas — and their facility taxonomies follow that function naturally. The chemical sector, by contrast, is defined by the substances it produces, moves, stores, and uses; those substances are embedded throughout the other critical-infrastructure sectors, and the regulatory and ownership structures that govern them are correspondingly fragmented.

A design basis is governed by the initiating events a facility must withstand and the mitigations available to answer them — not by the administrative sector to which a facility is assigned. Applying that placement principle to the chemical sector yields a functional decomposition rather than a substance-based one. This choice recovers the functional, consequence-based view of critical infrastructure that underlies dependency analysis in the NIPP: the sector list is an administrative and partnership construct, and the design basis follows the consequence chain rather than the administrative boundary. The same principle placed the hyperscale data center in

the electric sector's Critical Loads subsector by its consequence chain while leaving its NIPP sector of record as Information Technology.

Accordingly, DBA-CHEM is organized around four function classes — production (ProdCo), bulk transport (TransCo), distribution and supply (DisCo), and enterprise coordination and control (CorpCo) — with hazard class carried as a sub-index within each. This is the same relationship that the electric sector's generation function bears to its fuel-type sub-classes (coal, oil-fired, renewable). The decomposition and the sub-index are developed in Section 3.

1.4 Series Placement — DBA-CHEM as a Standalone Sector Series

DBA-CHEM is a standalone sector series. It is not a sub-sector of the Energy series and is not subordinate to any other sector framework; it is a peer of DBA-EN (Energy), DBA-W (Water), and the DBA-MA nuclear series. As with every sector series in the corpus, DBA-CHEM inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent framework and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination established as a cross-sector requirement in the DBA Series Gaps and Recommendations document. Inheritance of methodology is not subordination of sector: the chemical sector stands on its own consequence chains.

Section 2 — The Chemical Sector and Its Regulatory Context

2.1 A Multi-Regulator Sector

The chemical sector is governed by no single regulator. Process safety at fixed facilities is governed by the OSHA Process Safety Management standard (29 CFR 1910.119) and the EPA Risk Management Program rule (Clean Air Act §112r). Facility security was governed by the Chemical Facility Anti-Terrorism Standards (CFATS) administered by CISA until the program's statutory authority lapsed in July 2023, leaving a security-regulation gap that is itself a design-basis condition. Transport is governed by PHMSA and the Department of Transportation hazardous-materials regulations (49 CFR), with rail security governed by TSA directives. State fire codes, environmental agencies, and local emergency-planning committees add further layers. The regulator that applies depends on which function a facility performs.

2.2 Regulator-Per-Function Inheritance

Because the applicable regulator varies by function, the function-primary taxonomy is also the natural organizing structure for the sector's regulatory terrain. Each function class carries its own regulatory anchor: process-safety and risk-management regulation attaches at ProdCo; hazardous-materials and rail-security regulation attaches at TransCo; storage, fire-code, and facility-security regulation attaches at DisCo; and the enterprise information-technology and AI-governance obligations attach at CorpCo. What reads as regulatory fragmentation under a substance-based view becomes an orderly inheritance under a function-based one.

2.3 The Regulatory Gap

As in the water sector, no chemical-sector instrument today defines an engineering design basis — a formal statement of the events the infrastructure must be designed and demonstrated to withstand. Risk assessments, process-hazard analyses, and security vulnerability assessments exist; a design basis does not. The lapse of CFATS has widened the gap on the security dimension specifically. This framework constructs the design basis from existing consequence-analysis practice, on the premise that the consequences of chemical-sector failure — population-scale toxic release, and cascading loss of the chemical supplies on which other critical sectors depend — are severe enough to warrant design-basis thinking before an event compels it.

Section 3 — Function-Primary Taxonomy

3.1 The Four Function Classes

The chemical sector is decomposed into four function classes. Three are function classes describing the lifecycle of a chemical from creation to delivery, each realized as a physical facility class; the fourth is a coordination class describing the enterprise coordination and control layer, analogous in role to the electric sector's CntlCo but privatized into the owning firm rather than externalized into a sector authority. Each class is placed by the consequence chain it governs, carries its own regulatory anchor, and is sub-indexed by hazard class.

3.2 ProdCo — Production and Synthesis

The production function covers the manufacture and synthesis of chemicals. Its governing consequence is a toxic or flammable release — a source-term event — initiated by process upset, loss of containment, or compromise of the process-control system. Analytically this is the nearest neighbor of petroleum refining (DBA-OIL-FC1): the process-unit fire, explosion, or release is the bounding event, and long lead times for major process units make recovery a governing consideration. ProdCo is the node whose own inward hazard is highest.

3.3 TransCo — Bulk Transport

The transport function covers the bulk movement of chemicals by rail, pipeline, marine vessel, and truck. The toxic-inhalation-hazard (TIH) rail case — chlorine and anhydrous ammonia in particular — is the defining bounding event for this class, with a release consequence footprint set by population density along the transport corridor. This function shares a placement boundary with the Transportation sector: a TIH rail shipment is simultaneously a chemical-release object and a transportation object, and the boundary rule between DBA-CHEM-TR and any future Transportation-sector instrument must be stated where the two consequence chains meet. The Transportation Sector Framework states that rule (its §1.5, §3.4 and §6.3): the chemical-release chain is governed here and the transport-network chain by DBA-TRAN, with one shared release model, and DBA-CHEM-TR-FC2 (Bulk Chemical Transport) adopts it as written. Pipelines that

carry fuels stay with the Energy series; pipelines that carry chemicals other than fuels are governed by DBA-CHEM-TR-FC2 on the author's decision of 5 October 2026.

3.4 DisCo — Distribution, Blending, and Supply

The distribution function covers terminals, blending, packaging, and last-mile supply to end users — including the tank of chlorine or sodium hypochlorite that arrives at a water-treatment plant. Two properties distinguish this class. First, its inward hazard may be modest while its outward criticality is severe: a small repackaging or supply facility can be low-consequence on its own release potential yet carry a Tier-2 or Tier-3 designation because it is the sole supply path for a critical facility in another sector. Second, and consequently, the chemical sector's cross-sector criticality concentrates at the distribution boundary rather than at production — an inversion of the electric sector, where the highest-consequence facilities sit at generation and transmission. The elevation rule (Section 6) fires most often here. DBA-CHEM-DI-FC3 (Distribution, Blending and Supply Terminals) develops this: in that class either consequence chain may govern, and its governing military-action scenario is the denial of a sole supply rather than a release. On the author's decision of 5 October 2026 the class also takes agricultural chemical retailers holding ammonium nitrate or anhydrous ammonia.

3.5 CorpCo — Enterprise Coordination and Control

The chemical sector has no real-time shared commons and no sector-level reliability coordinator; unlike the electric grid, coordination is not externalized into an independent authority. It collapses instead into the owning enterprise, which functions as its own coordinator across its portfolio. CorpCo is that coordination class — the privatized analog of the electric sector's CntlCo. It is the locus at which the AI-Governance Design Basis is implemented: the Bright Line placement and the Command-Broker obligation for machine-learning and generative-AI systems in enterprise and operational-technology environments attach here, as does the analysis of portfolio-wide common-cause failure and of foreign-ownership risk — an axis this corpus already treats as design-basis-relevant for the hyperscale data-center class. DBA-CHEM-CorpCo (Chemical Enterprise Coordination and Control) develops this class, and states the Command Broker in full for a chemical company. On the author's decision of 5 October 2026 it also states one qualification of the placement here: when a hazardous shipment is in trouble, the Command Broker role for decisions about its movement is held within the carrier's control function under the Transportation series, and the shipping company owes its knowledge of the cargo to whoever holds a decision about the cargo itself. It is one role, held by different people for different actions. On the author's decision of 6 October 2026, the Command Broker's independence in this sector is procedural, as *The Broker* §4.5 allows where the deploying authority's interest is aligned with correct placement. The decisions in which a chemical company's commercial interest runs against refusal — deferring a shutdown to meet a delivery, running a sole supplier's stock down to save cost, restarting a fleet early after an event — are recorded in *Chemical Enterprise Coordination and Control* §6.5 rather than structurally separated. Recording them is

part of the justification from the domain's incentive structure that *The Broker* §4.5 requires, and the holder's ability to refuse without answering to the person who gains remains the safeguard.

† CorpCo is introduced here as a coordination-class designation mirroring the CntlCo coordination class in the electric sector. The designation was ratified 2026-08-09.

3.6 Hazard-Class Sub-Index and Hazard-Communication Standards

Substance is not fungible: chlorine, anhydrous ammonia, a reactive monomer, and a specialty polymer differ in hazard class, applicable regulation, and downstream dependents. Hazard class is therefore carried as a sub-index within each function class, in the same relationship that fuel type bears to the electric generation function.

The framework anchors that sub-index to the established regulatory taxonomy rather than a bespoke one. The formal key is the United Nations / U.S. Department of Transportation nine-class hazardous-materials system — explosives (1), gases (2), flammable liquids (3), flammable solids (4), oxidizers and organic peroxides (5), toxic and infectious substances (6), radioactive material (7), corrosives (8), and miscellaneous dangerous goods (9) — together with the four-digit UN/NA identification number, which is a machine-readable per-substance key, and its international sibling the Globally Harmonized System (GHS). Because the UN/DOT class is a transport-and-response classification rather than a consequence classification, the framework carries a consequence overlay on top of it: the poison- or toxic-inhalation-hazard (PIH/TIH) designation, persistence, and reactivity groupings that govern the design-basis consequence chain but are not fully captured by the class number alone.

The hazard-communication standards also attach to function, which independently corroborates the function-primary decomposition. The DOT diamond placard is an in-motion artifact carried on trucks, rail cars, tank cars, and intermodal containers, and belongs to TransCo; the NFPA 704 fire diamond is a fixed-facility artifact posted on buildings and storage tanks, and belongs to ProdCo and DisCo. The NFPA 704 zero-to-four ratings for health, flammability, and reactivity supply a ready severity scalar used in the consequence thresholds of Section 4.3. As the ProdCo, TransCo, and DisCo instruments mature, hazard classes that warrant their own bounding analysis will resolve into hazard-class-specific facility-class instruments keyed to the UN/DOT class.

Section 4 — The Chemical Sector Design Basis

[Skeleton — to be developed.] This section will define the envelope of adverse events the chemical sector must be demonstrated to withstand, organized by event category, with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion in the design envelope is consequence severity — potential for public-health impact at population scale, cascading failure across dependent CIKR sectors, or long-duration denial of a chemical supply on which a critical dependent facility relies — not event

likelihood. Events meeting the consequence threshold are in the envelope regardless of probability; probability belongs to risk management, not to the design basis.

4.2 Bounding Event Categories

[Stubs — each to be developed into a full event category with initiating event, consequence envelope, and required critical functions.] Candidate categories: toxic/flammable release from a production or storage unit; loss of containment of a TIH substance in transport; cyber compromise of process-control OT; supply-chain denial of a critical chemical to a dependent sector; and the chemical-release consequence chain of a precision strike on storage and feed systems.

Natural-hazard exceedance is a further category: a flood, hurricane, earthquake or extreme temperature beyond the hazard a facility's protective layers were designed for, which can remove a utility and its backups together. *Chemical Production and Synthesis* develops it at its §5.7, where each plant states its design-basis natural hazard (its §3.3), for flood with the source and date of the map recorded, and states what it does when the hazard exceeds that basis: the margin to the point at which every layer is lost, and the declared residual beyond it. Of the Arkema plant at Crosby, Texas, flooded by Hurricane Harvey in August 2017, the Chemical Safety Board found that it had no single flood design basis, that its safeguards could likely have protected it in a 100-year flood, and that the water stood more than two feet above the 500-year flood elevation; every layer of protection failed to the same flood, on the Board's finding (*Arkema Crosby Fire* F-1, F-3). The instrument for that event, DBA-CHEM-RE-2 (*Arkema Crosby Fire*), returns the category to this section at its R-3, and asks the production class to set the design-basis flood no lower than the most severe mapped floodplain, held in process safety information and renewed with each process hazard analysis cycle (*Arkema Crosby Fire* F-3, R-3). The category's development into a full event category, with the other candidates, remains open.

4.3 Consequence Thresholds and Critical Functions

The consequence thresholds are seeded from instrumentation the sector already maintains. The NFPA 704 health, flammability, and reactivity ratings give a first-order severity scalar per substance and facility; the Emergency Response Guidebook, keyed by the UN/NA number, supplies initial isolation and protective-action distances that serve as a first-order release-footprint proxy. The framework overlays on these the quantity on hand, the siting relative to population, and the cross-sector dependency — none of which the hazard-communication standards encode — to arrive at the design-basis consequence tier.

[Stub.] The critical functions whose continuity the design basis requires — containment, detection, isolation, and emergency notification — and the tier-assignment calculation to be developed here.

Emergency notification includes notification of the people within a release footprint. At Graniteville, South Carolina, on 6 January 2005, the volunteer fire chief took the first protective decisions within ten minutes without knowing the cargo; firefighters learned of the chlorine at twenty-seven minutes and the consist arrived by fax at forty-five; and the National Transportation Safety Board found that many if not all of the civilian deaths occurred before

responders could act. The instrument for that event, DBA-CHEM-RE-3 (*Graniteville Chlorine Release*), returns this element to the framework at its R-6. It returns the same element to *Bulk Chemical Transport* as a required function of that class's Chain R, with this critical function as its parent, and asks that class to state that the knowledge of the cargo it owes is owed within the plume's time, minutes, to the first responders and the public safety answering point, recording the Pipeline and Hazardous Materials Safety Administration's rule of June 2024 on real-time train consist information as the regulatory means (*Graniteville Chlorine Release* §6.6, F-6). The same instrument asks the transport class to test its required functions against the first ten minutes, in which, on its reading, the decisions that bore on the deaths at Graniteville were made (*Graniteville Chlorine Release* F-6). The tier-assignment calculation remains open.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Any chemical-sector facility may simultaneously carry a physical Design Basis established by the applicable DBA-CHEM instrument and an AI-Governance Design Basis governing the deployment of machine-learning and generative-AI systems in its operational and control environment. The two interact at the DBA level. Consistent with FD-BL, an action is above the Bright Line when its incorrect performance could produce a consequence at or above the facility's severity threshold and requires architecturally enforced authorization by a qualified Command Broker; determinism is a scope carve-out, not a placement rule. For the chemical sector the implementation locus of this architecture is CorpCo (Section 3.5), where enterprise and OT AI systems are governed.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Outward View — Chemical as Supply to Other Sectors

The forward-direction elevation obligation applies where a chemical-sector asset is the sole supply path for a facility carrying a Tier-2 or Tier-3 consequence designation in another DBA series. The governing examples are chlorine and sodium hypochlorite supply to water treatment (a Water-sector life-safety chain) and anhydrous ammonia supply to food and agriculture. In each case the supplying facility's design basis carries tightened acceptance criteria for that supply relationship. Because the supply hand-off occurs at the distribution boundary, the outward view is an attribute that fires the elevation rule at the DisCo node; it is not a separate facility class.

The four-digit UN/NA identification number is the machine-readable key for this linkage. A single identifier resolves a chemical — whether at a fixed facility or in transport — to its hazard class and, through it, to its dependent sectors: anhydrous ammonia (UN 1005) ties a distribution facility to both water disinfection and food-and-agriculture fertilizer supply at once, and chlorine

(UN 1017) ties it to municipal water treatment. Keying the elevation analysis to the UN number gives the outward view an auditable, machine-checkable hook.

6.2 Inward View — The Chemical Sector's Own Dependencies

The reverse-direction dependencies of the chemical sector run to energy (electric power and natural-gas feedstock and process heat), water (process and cooling water), and transport. A sustained loss of any of these is a design-basis condition for chemical-sector facilities where its consequence reaches the inclusion threshold, and the framework must state which energy and water assets are sole-supply paths for Tier-2 or Tier-3 chemical facilities so the reverse elevation can be applied by the supplying series.

6.3 The Same-Hazard / Different-Home Rule and the Shared Release Model

The identical physical hazard can be governed in more than one design-basis home depending on which facility's primary consequence chain dominates. Chlorine at a water-treatment plant is already addressed inside DBA-W-FC1, where the loss-of-disinfection chain dominates and the chemical-release chain rides along with it; the same chlorine at a bulk chemical terminal is governed by DBA-CHEM, where the release itself is the governing chain. To prevent divergent treatments of the same phenomenon, the chemical-release consequence model is developed once and cited by reference from every instrument that needs it — the propagate-by-reference discipline the corpus applies to all shared analysis. On the author's decision of 5 October 2026 the model is held provisionally at DBA-CHEM-PR-FC1 (Chemical Production and Synthesis) Section 4, and instruments cite it there until a designation is conferred on a standalone model instrument; when it moves, the instruments citing it are owed notice that a sweep is due.

Section 7 — Military-Action Extension Applicability

[Skeleton.] The MA extension is applied per facility under the mandatory two-factor determination — geolocation and strategic value — established as a cross-sector requirement in the DBA Series Gaps and Recommendations document (§1.7). For the chemical sector, strategic value is driven substantially by the outward-supply dependencies of Section 6: a distribution facility that is the sole chlorine supply for a metropolitan water system carries strategic value independent of its own size. The determination is time-varying and must be re-triggered on the defined geopolitical indicators.

7.1 Hazard-Communication as an Adversary Targeting Affordance

The hazard-communication standards that make the sector safer under accident conditions become an intelligence affordance under deliberate action. A DOT placard announces to any observer what a tank car carries; the UN/NA number on it resolves, through the publicly available Emergency Response Guidebook, to how the substance behaves and how far it reaches. Under the intentionality and precision characteristics of the military-action tier, mandated hazard transparency is reconnaissance the adversary is spared from performing — the same inversion the corpus identifies in data-center observability and in the Oldsmar detection-dependency case,

where a safety affordance reverses under deliberate action. The design-basis response cannot be to remove the information, because responders depend on it and the accident-domain benefit is real; it is to recognize hazard-communication exposure as a military-action design-basis condition and to answer it with controls that do not blind responders — routing and timing discipline, escort, real-time shipment tracking, and the rail-security posture TSA already governs. This finding is developed at the TransCo facility class and referenced at ProdCo and DisCo for their fixed-facility NFPA 704 analog.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-CHEM series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework. They do not repeat that material; each instrument adds what is specific to its function and hazard class — its bounding events, consequence chains, design parameters, and recovery architecture.

8.2 Facility Class Placement Rationale

Placement is function-primary and consequence-governed. ProdCo is registered first because production carries the highest inward release hazard; TransCo second, because bulk movement extends the release footprint along transport corridors; DisCo third, because it is the node at which cross-sector criticality concentrates through the elevation rule; and CorpCo as the coordination class carrying the enterprise coordination and AI-governance layer. Hazard class sub-indexes each entry and will resolve into dedicated instruments as the analysis matures.

8.3 Facility Class Register

The following table is the current facility-class register for the DBA-CHEM series. All four entries are at working-draft status, issued in October 2026: their inclusion establishes the planned scope of the series and the function-primary architecture; it does not represent completed analytical work. Designators were ratified 2026-08-09.

Designator	Function class (function-primary)	Hazard-class sub-index (UN/DOT-anchored)	Status	Primary regulatory anchor
DBA-CHEM-PR-FC1	ProdCo — chemical production & synthesis	UN/DOT classes 1–9 (esp. 2.3/6.1 PIH, 3, 5.1, 8); NFPA 704 overlay	Working draft	OSHA PSM (29 CFR 1910.119); EPA RMP (CAA §112r); CFATS (lapsed 2023 — open gap)
DBA-CHEM-TR-FC2	TransCo — bulk transport (rail · non-fuel chemical pipeline · marine · truck)	UN/DOT class via DOT placard; rail TIH (2.3/6.1)	Working draft	PHMSA/DOT (49 CFR); TSA (rail security directives)
DBA-CHEM-DI-FC3	DisCo — distribution, blending & supply terminals	by UN number & delivered product	Working draft	State fire/EPA storage; CFATS (lapsed 2023)

Designator	Function class (function-primary)	Hazard-class sub-index (UN/DOT-anchored)	Status	Primary regulatory anchor
DBA-CHEM-CorpCo	CorpCo — enterprise coordination & control (coordination class)†	n/a (enterprise / portfolio)	Working draft	Cross-cutting; AI-governance dual-DB (FD-BL / Command Broker)

Designators shown were ratified 2026-08-09 and confer no completed instrument. Facility-class numbers are sequential across the series; CorpCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo.

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-CHEM sector framework. It establishes the standalone-series placement, the function-primary taxonomy, the four function classes, the hazard-class sub-index, the regulator-per-function inheritance, and the cross-sector elevation structure. The design-basis envelope (Section 4), save the two elements Sections 4.2 and 4.3 take in, the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. DBA-CHEM-PR-FC1 (Chemical Production and Synthesis) has issued as the Phase-A instrument on the DBA-OIL-FC1 refinery-analog basis. It carries the shared chemical-release consequence model of Section 6.3 provisionally at its Section 4, and it proposes a custody-transfer boundary between production and bulk transport. DBA-CHEM-DI-FC3 (Distribution, Blending and Supply Terminals) has issued as the third, completing the series' physical classes; it adds detonation to the shared release model for stored ammonium nitrate, and it raises whether the question of who bears the cost of stricter supply criteria (OI-17) reaches the forward supply rule. DBA-CHEM-TR-FC2 (Bulk Chemical Transport) has issued as the second instrument, adopting the Chemical–Transportation boundary the Transportation Sector Framework states and placing non-fuel chemical pipelines in this series. Open items at this version: adopt or replace the custody-transfer boundary between production and bulk transport; confer a designation on the standalone release model when it is lifted out of DBA-CHEM-PR-FC1; take up the event categories, the independent-protection disposition step and the consequence-basis conditionality principle that DBA-CHEM-PR-FC1 developed ahead of this framework; take in, at a conformance pass of this framework, the elements that DBA-CHEM-RE-5 (*West Fertilizer Explosion*) and DBA-CHEM-RE-7 (*LockerGoga Attacks*) return to it, which *Distribution, Blending and Supply Terminals* §10.8 and *Chemical Enterprise Coordination and Control* §9.8 record, among them the notes of *LockerGoga Attacks* R-3 and R-6 for Section 2, the share of the separation distance held by the authority that permits building near a store (*West Fertilizer Explosion* F-2), and the holder of the disposal decision for material in a state's custody (*West Fertilizer Explosion* F-12); and write the fifth use case the author decided on 10 October 2026, on the TRITON material, to drive the governing cyber scenario of *Chemical Production and Synthesis* §6.4, which no use case drives now. Every reference-event instrument conferred at Section 9.1 is written. The first four returned two elements to this framework, which Sections 4.2 and 4.3 take in.

9.1 Conferred Reference Events

On the author's decision of 6 October 2026, this framework confers a reference-event designator on each of the seven events its four classes proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event on the model of the water series' DBA-W-RE-1 (*Minnesota Water Campaign*) and DBA-W-RE-2 (*Oldsmar Water Treatment*). All seven instruments are written. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are held by Project 11, DBA Reference Events; this framework keeps the conferral, and the six live chemical instruments are held there. On the author's triage ruling of 9 October 2026 one conferral, DBA-CHEM-RE-6, is withdrawn and its designator retired (below). The designators are numbered in the order the proposing classes were issued and, within each class, in the order that class lists its events. The number records conferral, not issue, and the instruments may be written in any order. Each designator is sector-primary and single: an event whose consequence reached other sectors is conferred here, in the sector that proposed it, and its reach is recorded in the register's impacted-sector column rather than in a second designator. That applies to four of the seven. The Graniteville and East Palestine events happened on railroads, which the Transportation series could claim; the Beirut explosion happened in a port; and the LockerGoga attacks also struck an aluminum producer, which belongs to Critical Manufacturing. Each is conferred here because its governing consequence is the chemical one.

From *Chemical Production and Synthesis*, two events are conferred: DBA-CHEM-RE-1, the TRITON attack of 2017 on the safety instrumented system of a petrochemical facility; and DBA-CHEM-RE-2, the Arkema fire at Crosby, Texas, on 31 August 2017. Their instruments are written: *The TRITON Attack on a Safety Instrumented System, 2017* and *The Arkema Crosby Fire, August 2017*, cited by the short titles *TRITON Attack* and *Arkema Crosby Fire*.

From *Bulk Chemical Transport*, two events are conferred: DBA-CHEM-RE-3, the collision and chlorine release at Graniteville, South Carolina, on 6 January 2005; and DBA-CHEM-RE-4, the derailment at East Palestine, Ohio, on 3 February 2023. Their instruments are written: *The Graniteville Chlorine Release, January 2005* and *The East Palestine Derailment, February 2023*, cited by the short titles *Graniteville Chlorine Release* and *East Palestine Derailment*.

From *Distribution, Blending and Supply Terminals*, two events are conferred: DBA-CHEM-RE-5, the explosion at West, Texas, on 17 April 2013; and DBA-CHEM-RE-6, the explosion at the port of Beirut on 4 August 2020. Their instruments were written. On the author's triage ruling of 9 October 2026 DBA-CHEM-RE-6 is withdrawn and its designator retired. Its two unique findings, that material held in state custody awaiting disposal belongs to no one's design basis and that the disposal decision had no holder, are carried into *The West Fertilizer Explosion, April 2013* as its F-11 and F-12, and *Distribution, Blending and Supply Terminals* and *Nodes and Terminals* (DBA-TRAN-ND-FC1) cite West in its place. The Beirut instrument stays with Project 9, in its archive, and does not move to Project 11. West is cited by the short title *West Fertilizer Explosion*.

From *Chemical Enterprise Coordination and Control*, one event is conferred: DBA-CHEM-RE-7, the LockerGoga ransomware attacks of March 2019 on Hexion and Momentive and on Norsk Hydro, read as a pair. Its instrument is written: *The LockerGoga Attacks, March 2019*, cited by the short title *LockerGoga Attacks*.

Every class states its events as conferred and written: *Chemical Production and Synthesis* and *Bulk Chemical Transport* at v0.4, and *Distribution, Blending and Supply Terminals* and *Chemical Enterprise Coordination and Control* at v0.3, each of which took in on 10 October 2026 the corrections and recommendations its events returned.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; FD-BR (The Broker); DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-OIL-FC1 (refinery release analog); DBA-W Sector Framework (chlorine/hypochlorite as named targets; §5.5 dual-DB model); DBA-W-RE-1 (Minnesota Water Campaign) and DBA-W-RE-2 (Oldsmar Water Treatment), the models for the reference-event instruments of Section 9.1; DBA-CHEM-RE-1 (TRITON Attack), DBA-CHEM-RE-2 (Arkema Crosby Fire), DBA-CHEM-RE-3 (Graniteville Chlorine Release), DBA-CHEM-RE-4 (East Palestine Derailment), DBA-CHEM-RE-5 (West Fertilizer Explosion) and DBA-CHEM-RE-7 (LockerGoga Attacks), the series' live reference-event instruments, held by Project 11 and cited by short title and finding or recommendation number; DBA-CHEM-RE-6 (Beirut Port Explosion) is withdrawn and kept by Project 9. Regulatory: OSHA 29 CFR 1910.119; EPA CAA §112r; PHMSA 49 CFR; TSA rail security directives; CFATS (lapsed July 2023).

Revision History

0.1 August 2026 — T. Roxey. Initial skeleton. Established DBA-CHEM as a standalone sector series; adopted the function-primary organizing principle (ProdCo / TransCo / DisCo / CorpCo) with hazard-class sub-index; set regulator-per-function inheritance; registered four Anticipated classes; framed the cross-sector elevation structure and the same-hazard/different-home rule. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.

0.2 August 2026 — T. Roxey. Anchored the hazard-class sub-index to the UN/DOT nine-class hazardous-materials system, with the four-digit UN/NA number as the machine-readable substance key and a PIH/TIH consequence overlay (§3.6). Mapped hazard-communication standards to functional classes — DOT placards to TransCo, NFPA 704 to ProdCo and DisCo. Seeded the consequence thresholds from NFPA 704 ratings and Emergency Response Guidebook isolation distances (§4.3). Added the hazard-communication-as-targeting-affordance finding to the military-action extension (§7.1). Keyed the cross-sector elevation to the UN number (§6.1).

0.3 August 2026 — T. Roxey. Designators ratified (PR / TR / DI + CorpCo); no code changes.

0.4 October 2026 — T. Roxey. Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier of ProdCo, TransCo, DisCo and CorpCo reads function class, and CorpCo as the enterprise coordination layer reads coordination class (OI-15). Section 5 already conformed to the October 2026 Edition and is unchanged; it is the model the other twelve frameworks were conformed to on 3 October 2026. No change to the design basis, the taxonomy or the register.

0.5 October 2026 — T. Roxey. DBA-CHEM-PR-FC1 (Chemical Production and Synthesis) issued at v0.1 and entered in the register at Section 8.3 as a working draft. On the author's decision of 5 October 2026, Section 6.3 names that instrument's Section 4 as the provisional home of the shared chemical-release consequence model (OI-18). Section 3.1 no longer calls ProdCo, TransCo and DisCo physical-function classes: they are function classes, each realized as a physical facility class under the author's determination of 30 September 2026, conformed on his ruling of 5 October 2026 carried by Project 0's handoff of that date. Section 9 records the issue and restates the open items it leaves. Revision-history entries are left as records. No change to the taxonomy, the designators, the hazard-class sub-index or the design basis.

0.6 October 2026 — T. Roxey. DBA-CHEM-TR-FC2 (Bulk Chemical Transport) issued and entered in the register at Section 8.3 as a working draft. Section 3.3 records that the Transportation Sector Framework states the Chemical–Transportation boundary rule and that DBA-CHEM-TR-FC2 adopts it, which corrects the 0.5 entry's Section 9, where the boundary was restated as still open although the Transportation framework had stated it; and it records the author's decision of 5 October 2026 placing non-fuel chemical pipelines in this series. Section 9 restates the open items, adding the division of Command Broker authority between carrier and shipper (OI-19). Register status cells no longer carry instrument version numbers, which belong to the instruments themselves. Revision-history entries are left as records. No change to the taxonomy, the designators, the hazard-class sub-index or the design basis.

0.7 October 2026 — T. Roxey. DBA-CHEM-DI-FC3 (Distribution, Blending and Supply Terminals) issued and entered in the register at Section 8.3 as a working draft, completing the series' three physical classes. Section 3.4 records what that instrument develops: either consequence chain may govern at distribution, supply denial is its governing military-action scenario, and agricultural chemical retailers holding ammonium nitrate or anhydrous ammonia belong to it on the author's decision of 5 October 2026. Section 9 records its issue, its addition of detonation to the shared release model, and the cost question it raises under OI-17. Revision-history entries are left as records. No change to the taxonomy, the designators, the hazard-class sub-index or the design basis.

0.8 October 2026 — T. Roxey. DBA-CHEM-CorpCo (Chemical Enterprise Coordination and Control) issued and entered in the register at Section 8.3 as a working draft, completing the series' four classes. Section 3.5 records the author's decision of 5 October 2026 dividing Command Broker authority between carrier and shipper when a hazardous shipment is in trouble, which closes OI-19, and names DBA-CHEM-CorpCo as the instrument that develops the class.

Section 9 replaces the OI-19 item with OI-20. Revision-history entries are left as records. No change to the taxonomy, the designators or the hazard-class sub-index.

0.9 October 2026 — T. Roxey. Section 9.1 confers DBA-CHEM-RE-1 through DBA-CHEM-RE-7 on the seven reference events the four classes proposed, on the author's decision of 6 October 2026, each committing the series to a register row and a standalone instrument, none yet written. They are numbered in the order the classes were issued and, within each class, in the order the class lists its events, and each is conferred in this sector, which proposed it, including the four whose consequence also reached Transportation or Critical Manufacturing. Section 3.5 records the author's decision of the same date that the Command Broker's independence in this sector is procedural, and that the commercially conflicted decisions named in Chemical Enterprise Coordination and Control §6.5 are recorded rather than structurally separated, which closes OI-20 here. Section 9 restates the open items: the OI-20 item is removed, and the item that named two reference events now names the seven instruments owed. References add FD-BR and the two water-series reference events. The facility-class instruments conform at their next revisions. Revision-history entries are left as records.

0.10 October 2026 — T. Roxey. Takes in the two elements that the series' first four reference-event instruments return to the framework, and records their issue. Section 4.2 names natural-hazard exceedance among the bounding event categories, with a design-basis natural hazard stated per facility, the margin to the loss of every layer and the declared residual beyond it, on DBA-CHEM-RE-2 (Arkema Crosby Fire) R-3, the Crosby flood having exceeded the 500-year elevation at a plant the Chemical Safety Board found had no single flood design basis. Section 4.3 states that emergency notification includes notification of the people within a release footprint, on DBA-CHEM-RE-3 (Graniteville Chlorine Release) R-6, and records that the transport class owes knowledge of the cargo within minutes to the first responders and the public safety answering point. The other recommendations of the four instruments return to the classes, the shared release model and the register, or go as notes to other series and to Layer Decomposition and Determinism, not to the framework. Section 9.1 records that four of the seven instruments are written, DBA-CHEM-RE-1 to DBA-CHEM-RE-4, that the production and transport classes state their events as conferred at v0.4, and that the instruments for DBA-CHEM-RE-5 to DBA-CHEM-RE-7 are owed, Chemical Enterprise Coordination and Control stating DBA-CHEM-RE-7 as conferred at v0.2; Section 9 restates the open item, and References add the four instruments. Section 4 otherwise remains a skeleton. An independent check against the four instruments raised 9 items, all taken before issue. Revision-history entries are left as records.

0.11 October 2026 — T. Roxey. Records the triage of the series' reference events, their move to Project 11, and the class conformance passes of 10 October 2026. Section 9.1 records all seven instruments as written and the six live ones as held by Project 11 under the determination of 9 October 2026, the conferral staying here, and replaces the statement, stale since 8 October 2026, that DBA-CHEM-RE-5 to RE-7 were owed. On the author's triage ruling of 9 October 2026 it

withdraws DBA-CHEM-RE-6 (the Beirut port explosion) and retires the designator, its F-1 and F-2 carried by DBA-CHEM-RE-5 (*West Fertilizer Explosion*) F-11 and F-12. Section 9.1 records that *Distribution, Blending and Supply Terminals* and *Chemical Enterprise Coordination and Control*, at v0.3, have taken in their events. Section 9 replaces the open item to write three instruments with the elements West and LockerGoga return to this framework, to be taken in at its next conformance pass, and records the fifth use case the author decided on 10 October 2026. The References list the six live instruments. An independent check of the revision against the notes, the instruments and the classes raised one item, the pending notice to Project 11, given the same day. Project 11 is given notice of this revision. Revision-history entries are left as records.